(ii) $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$.

Observe that $\frac{3\pi}{4}$ is not in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, the principal range of $\tan^{-1}x$.

So, we write $\frac{3\pi}{4} = \pi - \frac{\pi}{4}$.

Now, $\tan\left(\frac{3\pi}{4}\right) = \tan\left(\pi - \frac{\pi}{4}\right) = -\tan\frac{\pi}{4} = \tan\left(-\frac{\pi}{4}\right)$ and $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Hence, $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right) = -\frac{\pi}{4}$, since $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(iii) $\sec^{-1}\left(\sec\left(\frac{5\pi}{3}\right)\right)$.

Note that $\frac{5\pi}{3}$ is not in $[0, \pi]\setminus\left\{\frac{\pi}{2}\right\}$, the principal range of $\sec^{-1}x$.

We write $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$. Now, $\sec\left(\frac{5\pi}{3}\right) = \sec\left(2\pi - \frac{\pi}{3}\right) = \sec\left(\frac{\pi}{3}\right)$ and $\frac{\pi}{3} \in [0, \pi]\setminus\left\{\frac{\pi}{2}\right\}$.

Hence, $\sec^{-1}\left(\sec\left(\frac{5\pi}{3}\right)\right) = \sec^{-1}\left(\sec\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3}$.

(iv) $\sin^{-1}[\sin 10]$

We know that $\sin^{-1}(\sin\theta) = \theta$ if $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Considering the approximation $\frac{\pi}{2} \simeq \frac{11}{7}$,

we conclude that $10 \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, but $(10 - 3\pi) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Now, $\sin 10 = \sin\left(3\pi + (10 - 3\pi)\right) = \sin(\pi + (10 - 3\pi) = -\sin(10 - 3\pi) = \sin(3\pi - 10)$.

Hence, $\sin^{-1}[\sin 10] = \sin^{-1}\left[\sin\left(3\pi - 10\right)\right] = 3\pi - 10$, since $(3\pi - 10) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Example 4.18

Find the value of (i) $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$ (ii) $\cos\left[\frac{1}{2}\cos^{-1}\left(\frac{1}{8}\right)\right]$

(iii) $\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$.

Solution

(i) $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right] = \sin\left[\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right] = \sin\left(\frac{\pi}{2}\right) = 1$.

(ii) Consider $\cos\left[\frac{1}{2}\cos^{-1}\left(\frac{1}{8}\right)\right]$. Let $\cos^{-1}\left(\frac{1}{8}\right) = \theta$. Then, $\cos\theta = \frac{1}{8}$ and $\theta \in [0, \pi]$.

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